

From a fixed myogenesis gene set to an exploratory MYOD1 locus

Same-nucleus RNA + ATAC | COQ8A >=2 versus 1 UMI | TSS QC >=3 | 958 depth-matched pairs

A Define the search before looking at COQ8A effects

221 genes

Hallmark + Reactome



7,132 peaks

within 100 kb of a TSS



6,871 tested

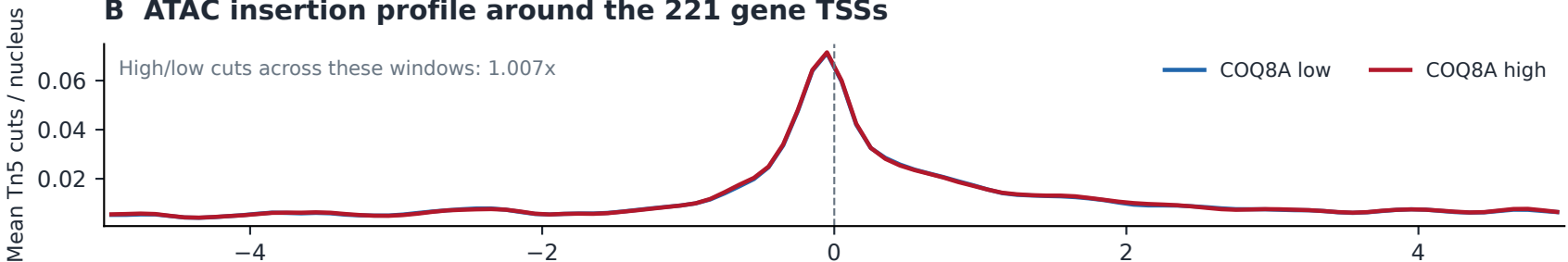
prevalence-qualified



0 FDR hits

primary ATAC screen

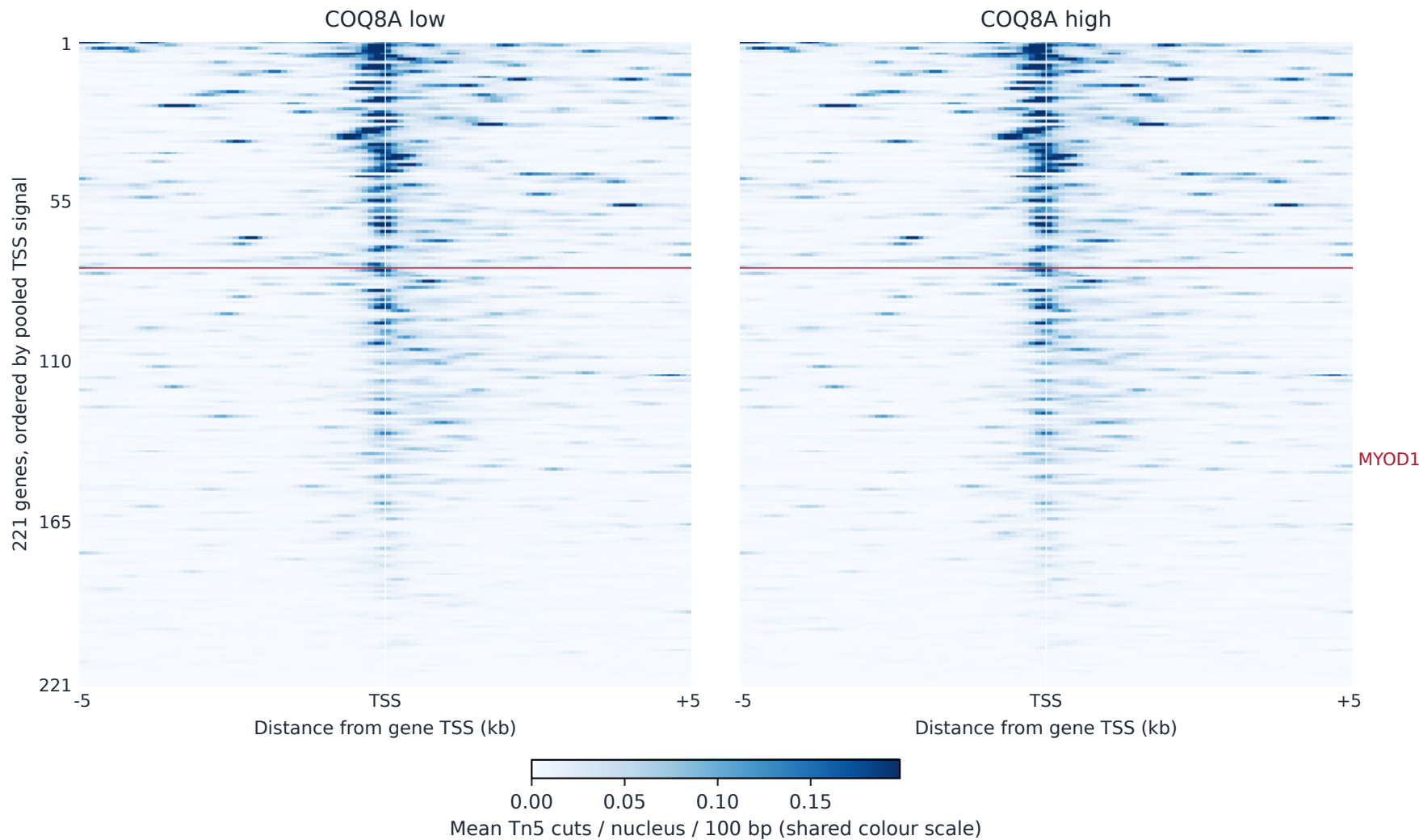
B ATAC insertion profile around the 221 gene TSSs



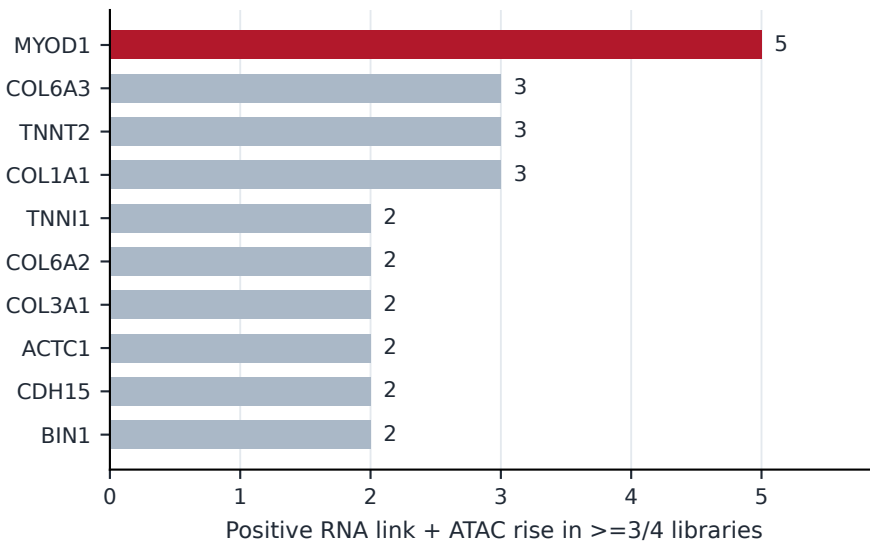
TSS heatmap: descriptive view

Locus ranking: all candidate peaks within 100 kb

MYOD1 six-peak ratio: a separate, exploratory subset



C Exploratory ranking of all 221 genes



D MYOD1: zoom from 32 regions to 6

32 nearby -> 19 shared -> 11 linked (TSS QC >=2) -> 6 motif-low

